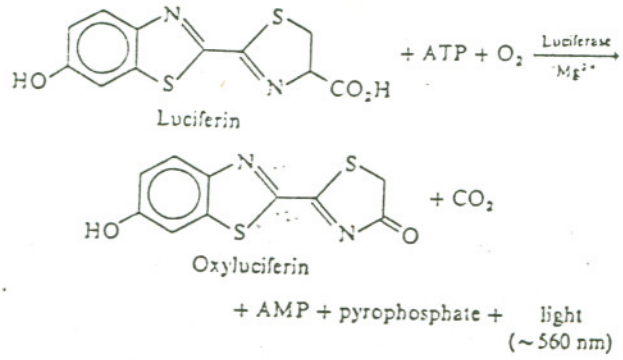
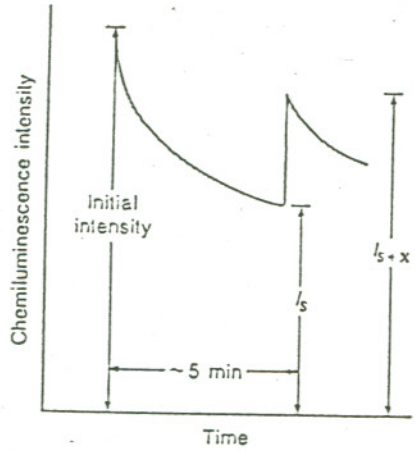
ANALYTICAL CHEMISTRY CUMULATIVE EXAM

December 7, 1991

A sensitive assay for ATP is based on its participation in the light-producing reaction of the firefly. The reaction catalyzed by the enzyme luciferase is



When the reactants are mixed, the solution gives off light. The light intensity decays slowly due to product inhibition of the reaction. Otherwise, the light would have a steady intensity because *the rate at which reactants are consumed is negligible.* That is, ATP and luciferin maintain their original concentrations throughout the few minutes that the reaction might be monitored. Some typical experimental results are shown below.



Let the initial concentration of ATP in the reaction be [S]. Suppose that additional ATP is added, increasing the concentration in the reaction to [S] + [X]. The kinetic description of the reaction predicts that the increase in light intensity after the addition will be given by

$$\frac{I_s}{I_{s+x}} = \frac{1}{[S]+[X]}\left(\frac{K[S]}{K+[S]}\right) + \frac{[S]}{K+[S]}$$

where K is constant.

(a) Suppose that [S] = 250 μM and after 5 minutes I_S = 58.7 arbitrary intensity units. Then a standard addition of [X] = 200 μM is made, and I_{S+X} is found to be 74.5 units. Use these data to find the value of K in the equation above.

(b) A short time later, when the intensity had decayed to 63.5 units, an unknown aliquot of ATP was added to the reaction, and the intensity increased to 74.6 units. Find [X], the concentration increase in the reaction vessel caused by the ATP aliquot.

(c) Explain what you would have to know, and do, to find the concentration of ATP <u>in</u> the aliquot.

TABLE OF NUCLIDES

Z	El	A	Abundance or $t_{1/2}$	Decay Mode	Δ(MeV)	Jπ	σ_n(b)
0	n	1	10.6 m	β^-,no γ	8.071	1/2+	
1	H	1	*99.985%*		7.289	1/2+	0.332
		2	*0.0148%*		13.136	1+	5.2×10^{-4}
		3	12.33 y	β^-,no γ	14.950	1/2+	$<6\times10^{-6}$ sc
2	He	3	*1.38×10^{-4}%*		14.931	1/2+	5.33×10^{3} np
		4	*99.99986%*		2.425	0+	
		6	0.808 s	β^-,no γ	17.597	0+	
		8	0.122 s	β^-,β^-n 12%	31.609	0+	
3	Li	6	*7.5%*		14.087	1+	942 na
		7	*92.5%*		14.908	3/2−	0.045 rs
		8	0.84 s	$\beta^-2\alpha$	20.947	2+	
		9	0.178 s	β^-,β^-n2α 35%	24.955	(3/2)−	
		11	8.5 ms	β^-,β^-n 61%	40.94		
4	Be	7	53.3 d	EC	15.770	3/2−	5×10^{4} rs np
		9	*100%*		11.348	3/2−	0.008 np
		10	1.6×10^{6} y	β^-,no γ	12.608	0+	<0.001 rs
		11	13.8 s	β^-,$\beta^-\alpha$ 3%	20.176	1/2+	
		12	11.4 ms	β^-,β^-n	25.03	0+	
5	B	8	0.769 s	$\beta^+2\alpha$	22.922	2+	
		10	*19.8%*		12.052	3+	3838 na
		11	*80.2%*		8.668	3/2−	0.005 rs
		12	20.4 ms	β^-,β^-3α 1.6%	13.370	1+	
		13	17.4 ms	β^-,β^-n 0.28%	16.562	3/2−	
		14	16 ms	β^-	23.657	2−	
6	C	9	0.1265 s	β^+p2α	28.912	(3/2−)	
		10	19.2 s	β^+	15.703	0+	
		11	20.38 m	β^+ 99.76%, EC 0.24%,no γ	10.650	3/2−	
		12	*98.89%*		0	0+	0.0034
		13	*1.11%*		3.125	1/2−	9×10^{-4}
		14	5730 y	β^-,no γ	3.020	0+	$<1\times10^{-6}$ rs
		15	2.449 s	β^-	9.873	1/2+	
		16	0.75 s	β^-n >98.8%	13.693	0+	
7	N	12	11.0 ms	β^+,β^+3α 3.5%	17.338	1+	
		13	9.96 m	β^+,no γ	5.346	1/2−	
		14	*99.63%*		2.863	1+	1.82 sc np
		15	*0.366%*		0.102	1/2−	4×10^{-5} rs
		16	7.13 s	β^-,$\beta^-\alpha$ 0.0012%	5.682	2−	
		17	4.17 s	β^-,β^-n 95%	7.870	1/2−	
		18	0.63 s	β^-	13.274	0,1,2−	
8	O	13	8.9 ms	β^+p	23.105	3/2−	
		14	70.60 s	β^+	8.008	0+	
		15	122 s	β^+ 99.89%, EC 0.11%,no γ	2.855	1/2−	
		16	*99.76%*		−4.737	0+	1.8×10^{-4} rs
		17	*0.038%*		−0.810	5/2+	0.235 sc na
		18	*0.204%*		−0.783	0+	1.6×10^{-4}
		19	26.9 s	β^-	3.331	5/2+	
		20	13.5 s	β^-	3.799	0+	
9	F	17	64.5 s	β^+,no γ	1.952	5/2+	
		18	109.8 m	β^+ 96.9%, EC 3.1%,no γ	0.872	1+	
		19	*100%*		−1.487	1/2+	0.010 rs
		20	11.0 s	β^-	−0.017	2+	
		21	4.32 s	β^-	−0.047	5/2+	
		22	4.23 s	β^-	2.826	4+	
		23	2.2 s	β^-	3.35	(5/2)+	
10	Ne	17	0.109 s	β^+p	16.478	1/2−	
		18	1.67 s	β^+	5.319	0+	
		19	17.3 s	β^+ 99+%,EC 0.102%	1.751	1/2+	
		20	*90.51%*		−7.043	0+	0.038 rs
		21	*0.27%*		−5.733	3/2+	0.7 rs

Avogadro's No = 6.022×10^{23}